

AI-Driven Code Review Architecture

1. Overview

This document describes a modular, scalable AI-assisted code review architecture designed for enterprise use. It integrates static analysis, machine learning models, rules engines, and automated feedback workflows.

2. Components

- Code Intake Layer

Accepts source code via API, CI/CD hooks, or manual uploads.

- Pre-Processing Engine

Normalizes code, extracts metadata, identifies language/framework, and prepares embeddings.

- Static Analysis Module

Performs linting, security scans, dependency checks, and complexity analysis.

- AI Review Engine

Includes:

- Large Language Model evaluator
- Code smell predictor
- Security heuristic engine
- Auto-fix suggestion generator

- Rule-Based Policy Engine

Enforces organization-specific constraints and quality gates.

- Feedback Aggregator

Merges AI findings + static rules into a single structured response.

- Reporting & Exporter

Outputs feedback as JSON, PDF, or integrates with GitHub, GitLab, Azure DevOps.

3. Data Flow

Developer → Intake API → Pre-Processing → Static Analysis + AI Model → Policy Engine → Aggregated Review → Report

4. Benefits

- Faster review cycles
- Consistency across reviewers
- Automated detection of common defects
- Support for multiple languages and frameworks
- Extensible for enterprise compliance rules